

ASHOK GURUNG, Ph.D.

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Summary

Computational physicist with 6+ years of experience in multiscale and multiphysics modeling of electro-chemo-mechanical phenomena in energy materials. Expertise spans phase-field modeling, finite element methods, molecular dynamics, and density functional theory for solid-state batteries and functional ceramics. Experienced in integrating simulations with experiments and leveraging Python-based automation, high-performance computing, and data-driven/AI-assisted approaches for predictive materials design.

Education

University of Connecticut (UConn), CT, US

Ph.D. in Physics, Dec 2024

Wright State University (WSU), OH, US

M.S. in Physics, July 2018

Technical Skills

Computational techniques: Finite Element Modeling (FEM) via MOOSE; COMSOL; Classical Molecular Dynamics; Density Functional Theory (DFT) via VASP; Data Mining; Machine Learning

Programming and Software: Python; MATLAB; C/C++; shell script; MS Office; LaTeX; GitHub

Operating System: Ubuntu; high-performance computing cluster; and Windows OS

Other Skills: Interpersonal & Leadership Skills; Scientific Writing & Presentation Skills; Project Management & Organization Skills; Time Management Skill; Teamwork Skill; Independent Work Skill

Publications

- A. N. Bage, J. M. Vazquez Mercado, F. D. Cúñez, **A. Gurung**, S. Li and H. Tu, “*Dendrite Suppression by Detouring Li Transport within a Mechanically Anisotropic Solid Electrolyte*”, Nano Letters (2026) .
- R. Elawadly, J. M. Vazquez Mercado, F. D. Cúñez, **A. Gurung**, I. A. Shozib, Q. H. Tu, *Tuning Effective Mechanical Properties of Catholyte to Enhance Cathode Performance in Solid-state Batteries*, 2025, <http://dx.doi.org/10.2139/ssrn.5534041>
- **A. Gurung**, M. Fatin Ishtiyag, T. Sahoo, S. M. Nakhmanson, S. K. Nayak, S. P. Alpay, and P. E. Janolin, “*Electrostriction of Organic Compounds*” (in preparation 2025).
- **A. Gurung**, J. M. Mangeri, M. Fatin Ishtiyag, S. P. Alpay, and S. Nakhmanson, “*Extrinsic Dielectric Response due to Domain Wall Motion in Ferroelectric BaTiO₃*”, Computational Materials Today. 5, 100016 (2025).
- **A. Gurung**, J. M. Mangeri, A. M. Hagerstrom, N. D. Orloff, S. P. Alpay, and S. Nakhmanson, “*Modeling Structure-Properties Relations in Compositionally Disordered Relaxor Dielectrics at the Nanoscale*”, J. Appl. Phys. 134, 104102 (2023).
- **A. Gurung***, D. Trujillo* (equal contribution), J. Yu, S. K. Nayak, S. P. Alpay, and P-E. Janolin, “*Data-driven Methods for Discovery of Next-generation Electrostrictive Materials*”, npj Comp. Mater. 8, 251 (2022).
- **A. Gurung**, “*Molecular Dynamics Study of Hydrogen Trapping and Helium Clustering in Tungsten*”, master’s thesis, Wright State University, 2018.

Professional Experience

Phase field modeling of sintering of ceramic materials

Jun 2025 – Present

Postdoctoral Research Associate, School of Mechanical and Materials Engineering, WSU

Pullman, WA

- Developed a coupled phase-field and concentration-field formulation to model sintering mechanisms, including surface, bulk, and interface diffusion, enabling quantitative analysis of microstructural evolution.
- Integrated mechanical modulus effects into the model to simulate sintering behavior under various experimental setups, providing insights that guided experimental parameter selection.

- Investigated temperature-dependent sintering processes to identify conditions influencing ceramic densification, thereby predicting resulting changes in optical and mechanical properties.
- Implemented a robust numerical scheme for joining dissimilar materials at elevated temperatures, supporting design of high-performance composite interfaces.

NSF-Engine Workforce Development Summer Camp

Apr 2025 – May 2025

Postdoctoral Research Associate, Department of Mechanical Engineering, RIT

Rochester, NY

- Participated in curriculum development for the 2025 NSF-Engine Workforce Development Summer Camp for community college students.
- Designed and delivered a lecture on galvanic cells and their components, adapting content to accessibility to students without prior technical background.
- Developed instructional materials explaining battery classification (based on life cycle, format, electrode architecture, etc.) and operational principles.
- Collaborated with graduate students Radwa Elawadly and Imtiaz Shozib to integrate lecture content with hands-on experiments, ensuring theoretical concepts were reinforced through experiential learning.

Multiphysics Modeling for solid state battery

Sep 2024 – May 2025

Postdoctoral Research Associate, Department of Mechanical Engineering, RIT

Rochester, NY

- Developed and implemented a multiscale and multiphysics electro-chemo-mechanical framework (MOOSE + DEM) to quantify mechanically induced anisotropy in uniaxially pressed solid electrolytes, revealing direction-dependent Li transport and dendrite propagation mechanisms.
- Designed and experimentally validated an asymmetric ring-electrode architecture to redirect Li flux toward mechanically robust orientations (45°), increasing critical current density from 0.9 to 5.47 mA/cm² ($5\times$ enhancement).
- Established a quantitative relationship between catholyte effective mechanical properties (elastic modulus, compressive strength, toughness) and interfacial contact stability in composite cathodes using coupled continuum–granular simulations.
- Led microstructural engineering of sulfide catholytes via particle-size refinement (ball milling), reducing effective stiffness while preserving strength and improving electro-chemo-mechanical stability and full-cell capacity retention in ASSBs.

Quantum Field Expert and Outreach Specialist

Sep 2024 – Nov 2024

Postdoctoral Research Associate, Department of Mechanical Engineering, RIT

Rochester, NY

- Contributed as a quantum field expert at the event titled "Lecture: What Are Quantum Technologies?", organized by Quantum CT, a collaborative initiative between the University of Connecticut and Yale University.
- Played a key role in promoting quantum technologies and their impact on economic growth and societal advancement.
- Prepared and tailored content to educate diverse audiences, including educators, businesses, policymakers, and engineers, on the importance of quantum technologies and their future impact.
- For more details, visit: <https://quantumct.org/events/quantum-up-challenge/panel-sessions/>

Mesosopic and Continuum Scale Modeling

Aug 2020 – Aug 2024

Research Assistant–Funded by Air Force Research Lab (AFRL) and Industry

Storrs, CT

- Explored diverse ferroelectric systems using Time Dependent-Landau Ginzburg Devonshire (TD-LGD) approach and phase-field modeling.
- Developed and applied physics based computational tool (in house built finite element software (based on C/C++) to simulate structure-property relationships in ferroelectrics materials.

- Modeled dielectric dispersion response in different ferroelectric systems utilizing an original approach developed at UConn and collaborating institutions.
- Installed and maintained an open-source finite element Multiphysics Object-Oriented Simulation Environment (MOOSE) and in-house built software–Ferret; performed calculations using an HPC environment.
- Extensively leveraged Python packages for both pre- and post-processing stages, developed and optimized high performance data workflow algorithm to perform efficient extraction and subsequent analysis of simulated data.
- Performed thorough statistical sampling and analysis across various post-processing stages, handling extensive datasets.
- Explored magnetic systems at the mesoscale using the Landau-Lifshitz-Gilbert (LLG) model.
- Developed, tested, validated, and applied the micromagnetic module developed in Ferret to study the properties of magnetic system.
- Developed and explored functional properties, including electro-optical, magneto-optical, and elasto-optical responses in optical materials, utilizing Multiphysics coupling approach.
- Participated in the development of a continuum-level solidification model. Analyzed and predicted thermal evolution of a system during solid-liquid phase transformation.
- Have experience in effectively working in a team and collaborating on a project spanning diverse research backgrounds, including atomistic modeling, metamaterial design, device fabrication, material growth, and characterization.
- Possess strong communication skills with extensive experience in creating presentations, delivering them at conferences, and authoring peer-reviewed journal papers.

Atomistic modeling–(Density Functional Theory)

Aug 2018 – Aug 2020

Research Assistant, Physics Department, UConn

Storrs, CT

- Implemented a data-driven approach to derive the proxy formula for the electrostrictive tensor using thermodynamic potential.
- High throughput screening of materials from the Material Project database and employed Density Functional Theory (DFT) for robust calculations to identify materials with high electrostriction coefficients.

Atomistic Modeling–(Classical Molecular Dynamics)

Aug 2016 – July 2018

Research Assistant, Physics Department, WSU

Dayton, OH

- Employed classical molecular dynamics (MD) simulations to investigate the impact of hydrogen energy and surface temperature on hydrogen retention in a tungsten matrix within a nuclear fusion reactor.
- Focused on predicting the safety implications of hydrogen isotope accumulation, particularly tritium, for the overall reactor performance and operation.
- Investigated helium bubble formation and its influence on hydrogen trapping in a tungsten matrix at various surface orientations, aiming to predict and understand the durability and thermoelectric conversion efficiency of fusion reactors.

Presentations (Oral) and Conferences

- “*Modeling the dielectric dispersion of BaTiO₃ with a local phase field method*”, Materials Science and Technology Conference, October 2023, Columbus OH, USA
- “*Modeling the relaxor dielectric dispersion of Ba_{1-x}Sr_xTiO₃ with a local phase field method*”, Materials Science and Technology Conference, October 2022, Pittsburgh PA, USA

- “*Modeling the relaxor dielectric dispersion of with a local phase field method*”, Electronic Materials and Applications Conference, January 2022, Virtual participation, USA
- “*Predicting dielectric response in 5G materials with FEM simulations*”, Industrial affiliates program (IAP) annual meeting, July 2019, UConn, Storrs CT, USA.

Professional Memberships and Services

- Served as a peer reviewer for Physical Review B, Physical Review Applied, ACS Applied Materials & Interfaces, Discover Materials, Physical Review journals, Reviews of Modern Physics
- American Physical Society
- American Ceramic Society (ACerS)
- Material Advantage
- Association for Iron and Steel Technology (AIST)

Workshops

- “*Moving ions with VASP*” Organized by VASP, Virtual, Nov 2023.
- “*MOOSE Workshop*” Organized by Idaho National Lab, Virtual, June 2023.
- “*Quantum Matter: Dynamics and Sensor*” jointly organized by the University of Connecticut, University of North Florida, and NORDITA, December 2022.
- “*CHiMaD Phase Field Workshop XIV*” organized by Center for Hierarchical Material Design, Evanston, IL, Hybrid, October 2022.
- “*CHiMaD Phase Field Workshop XII*” organized by Center for Hierarchical Material Design, Virtual, November 2021.
- “*IMS Quantum Matter Group 2019*” held at Los Alamos National Lab (LANL) as a visiting graduate student for the summer program 2019.

Honors Scholarships and Fellowships

- “*Modeling structure-properties relations in compositionally disordered relaxor dielectrics at the nanoscale*” paper **featured on the cover of Journal of Applied Physics** (2023).
- Conference Participation Award, UConn (2023).
- Doctoral Dissertation Fellowship, UConn (2023).
- Pre-Doctoral Dissertation Fellowship, UConn (2019).
- Guest Travel Fund, Los Alamos National Lab (LANL) (2019).

Leadership

- Senate, Graduate Student Senate, (2021-2023)
- Secretary, Nepalese Graduate Student Association, (2021-2022)
- Member, Student Welfare Committee, (2022-2023)
- Member, Graduate Employee and Postdoc Union (2018-2023)

References

- Dr. Serge Nakhmanson, serge.nakhmanson@uconn.edu, Associate Professor, Department of Materials Science and Engineering, University of Connecticut, CT, USA
- Dr. Scott Beckman, scott.beckman@wsu.edu, Professor, Acting Director, Materials Science and Engineering Program, Washington State University, WA, USA
- Dr. Sinisa Mesarovic, smesarovic@wsu.edu, Professor, Materials Science and Engineering Program, Washington State University, WA, USA
- Dr. Pierre-Eymeric Janolin, pierre-eymeric.janolin@centralesupelec.fr, Professor, CentraleSupélec, Université Paris-Saclay, Gif-sur-Yvette, France